Casual Document for Quantitative Mixing Time Results for Single Site Gibbs in Stochastic Block Model

Wen, Zehai; Cesare, Michael

2026-11-09 15:32:28-05:00

Contents

1 The Stochastic Block Model and the Single Site Gibbs Method **2**

Chapter 1

The Stochastic Block Model and the Single Site Gibbs Method

1.1 Basic Setup

The procedure of analyzing graphical data using Bayesian stochastic block model with single site Gibbs procedure seems to have been first mentioned in Kemp et al. (2004) and slightly later in Xu et al. (2006). In this section, we describe the simplest setup of both the model and the single site Gibbs method by following a tutorial article Schmidt and Mørup (2013).

For the moment, let us consider the simplest case where the graphical data received by the statistician is just one simple graph of $N \geqslant 3$ vertices labeled as $1, 2, \ldots, N$, represented by its adjacency matrix $X \in \{0,1\}^{N \times N}$ where $X_{i,j} = 1$ if and only if there is an edge between vertex i and vertex j. In particular, we have $X_{i,i} = 0$ for all i and $X_{i,j} = X_{j,i}$ for all i, j. If we want to, we can consider later the case which our data contains multiple graphs of the same size generated independently identically distributed from a distribution in the same model, but we shall not do so in this section.

We assume that X is generated from a distribution in the (standard Bayesian) stochastic block model (SBM) with unknown number of blocks. To describe the model, let us write Π for the collection of all partitions of the set $\{1, 2, \ldots, N\}$ and represent any element $z \in \Pi$ using the *restricted growth string* notation, meaning that we see z as a vector $z = (z_1, z_2, \ldots, z_N)$ where z_i denotes the block to which vertex i belongs, such that the classes are ordered as they first appear: $z_1 = 1$ and, for every $i \in \{2, 3, \ldots, N\}$, we have

$$z_i \in \left\{1, 2, \ldots, 1 + \max_{j<i} z_j\right\}.$$

For example, if $N = 5$ and the partition is $z = \{\{1, 3\}, \{2\}, \{4, 5\}\}$, then we write $z = (1, 2, 1, 3, 3)$. As derived in Schmidt and Mørup (2013), (the standard Bayesian) SBM is a parameteric model for the matrix X whose only parameter is z, ranged over Π, with density function is given by

$$p(X \mid z) = \prod_{1 \leqslant k \leqslant l \leqslant K} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a, b)}, \tag{1.1}$$

where $a > 0$ and $b > 0$ are hyperparameters fixed by the statistician, $B(\cdot, \cdot)$ is the beta function, K is the number of non-empty blocks as specified by z, $m_{k,l}$ is the number of edges between block k and block l, and $\bar{m}_{k,l}$ is the number of non-edges between block k and block l.

(Actually, just as in Schmidt and Mørup (2013), we have abused the notation a bit in (1.1). Given that X is symmetric with diagonal entries equal to 0, we are only interested in the vector $(X_{i,j})_{1 \leqslant i < j \leqslant N}$ and we have written $p(X \mid z)$ in place of $p((X_{i,j})_{1 \leqslant i < j \leqslant N} \mid z)$.)

In order to do Bayesian statistics, we have to specify a prior distribution for z. As pointed out in Schmidt and Mørup (2013), we will use the Chinese restaurant process (CRP) prior for z. With a hyperparameter $A > 0$

fixed by the statistician, CRP(A) is a probability measure on Π defined as follows: $z_1 \equiv 1$ and, for every $i \in \{2, 3, \ldots, N\}$, given $z_1, z_2, \ldots, z_{i-1}$ who have already used a total of $K^{(i-1)}$ blocks, z_i will join an existing block $k \in \{1, 2, \ldots, K^{(i-1)}\}$ with probability

$$\frac{n_k^{(i-1)}}{A + i - 1},$$

where $n_k^{(i-1)}$ is the number of vertices among $1, 2, \ldots, i-1$ who have been assigned to block k, and will create a new block $K^{(i-1)} + 1$ with probability

$$\frac{A}{A + i - 1}.$$

CRP(A) does have a clean joint density (with respect to the counting measure on Π)

$$p(z) = \frac{A^K \Gamma(A)}{\Gamma(A + N)} \prod_{k=1}^{K} \Gamma(n_k), \tag{1.2}$$

where K is the number of non-empty blocks as specified by z, n_k is the number of vertices assigned to block k according to z, and $\Gamma(\cdot)$ is the gamma function.

By Bayes's theorem, according to (1.1) and (1.2), the posterior distribution of z given X is

$$p(z \mid X) \propto p(X \mid z)p(z) = \left[\prod_{1 \leqslant k \leqslant l \leqslant K} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a, b)} \right] \left[\frac{A^K \Gamma(A)}{\Gamma(A + N)} \prod_{k=1}^{K} \Gamma(n_k) \right].$$

We now move to carry out the single site Gibbs method. For every $i \in \{1, 2, \ldots, N\}$, let $z^{(-i)}$ denote the vector $(z_1, z_2, \ldots, z_{i-1}, z_{i+1}, \ldots, z_N)$ obtained from z by removing the i-th entry. We need to compute, for every $i \in \{1, 2, \ldots, N\}$, what's proportional to

$$p(z_i \mid z^{(-i)}, X).$$

Therefore, for each $k \in \{1, 2, \ldots, K^{(-i)}\}$, where $K^{(-i)}$ is the number of non-empty blocks as specified by $z^{(-i)}$, we must calculate $p(z_i = k \mid z^{(-i)}, X)$.

1.2 Gibbs Draft

In this section we explain how we understand the conditional distribution $p(z_i \mid z^{(-i)}, X)$ used by the single site Gibbs method, given that z ranges over the set of partitions Π rather than over a product space such as $\mathbb{R}^N$. Throughout the section, we fix a vertex $i \in \{1, 2, \ldots, N\}$ and we regard every $z \in \Pi$ as a set of blocks, each block being a non-empty subset of $\{1, 2, \ldots, N\}$. The restricted growth string notation is only used to write down a partition after it has been updated; it plays no role in the update itself.

The partition of the other vertices and where vertex i goes

Let $\Pi^{(-i)}$ denote the collection of all partitions of the set $\{1, 2, \ldots, N\} \setminus \{i\}$. Every $z \in \Pi$ induces a partition of the other $N - 1$ vertices, namely its restriction

$$T_i(z) = \{C \setminus \{i\} : C \in z\} \setminus \{\varnothing\} \in \Pi^{(-i)}.$$

For $\rho \in \Pi^{(-i)}$ with K_ρ blocks $B_1, B_2, \ldots, B_{K_\rho}$, which we name by their vertex sets rather than by integer labels, let

$$\mathcal{C}_\rho = \{B_1, B_2, \ldots, B_{K_\rho}, \text{new}\}$$

be the set of the $K_\rho + 1$ options for vertex i: join one of the existing blocks of ρ, or start a new block. For $c \in \mathcal{C}_\rho$, we write $\rho \oplus_i c \in \Pi$ for the partition obtained from ρ by putting i into c, that is,

$$\rho \oplus_i B_k = (\rho \setminus \{B_k\}) \cup \{B_k \cup \{i\}\} \quad \text{for } k \in \{1, 2, \ldots, K_\rho\}, \qquad \rho \oplus_i \text{new} = \rho \cup \{\{i\}\}.$$

Lemma 1. The map $(\rho, c) \mapsto \rho \oplus_i c$ is a bijection from $\{(\rho, c) : \rho \in \Pi^{(-i)},\ c \in \mathcal{C}_\rho\}$ onto Π. Its inverse sends $z \in \Pi$ to $(T_i(z), c_i(z))$, where, writing C for the block of z containing i, we set $c_i(z) = C \setminus \{i\}$ if $C \neq \{i\}$ and $c_i(z) = \text{new}$ if $C = \{i\}$.

Proof. For every $\rho \in \Pi^{(-i)}$ and $c \in \mathcal{C}_\rho$, removing i from $\rho \oplus_i c$ gives back ρ, and the block of $\rho \oplus_i c$ containing i is $c \cup \{i\}$ if $c \neq \text{new}$ and $\{i\}$ if $c = \text{new}$, so $T_i(\rho \oplus_i c) = \rho$ and $c_i(\rho \oplus_i c) = c$. Conversely, for every $z \in \Pi$, putting i back into $c_i(z)$ recovers z, so $T_i(z) \oplus_i c_i(z) = z$. □

In the coordinates given by Lemma 1, $\rho = T_i(z)$ plays the role of $z^{(-i)}$ and $c = c_i(z)$ plays the role of z_i. Accordingly, for $\rho \in \Pi^{(-i)}$ and $c \in \mathcal{C}_\rho$, we define

$$p(z_i = c \mid z^{(-i)} = \rho, X) = \frac{p(\rho \oplus_i c \mid X)}{\sum_{c' \in \mathcal{C}_\rho} p(\rho \oplus_i c' \mid X)} \propto p(X \mid \rho \oplus_i c)\, p(\rho \oplus_i c). \tag{1.3}$$

This is a probability distribution over the $K_\rho + 1$ options in $\mathcal{C}_\rho$, not over integer labels: no block label appears anywhere. If i forms a block on its own in the current state z, then $T_i(z)$ has one block fewer than z and the current state corresponds to the option $c = \text{new}$; this case needs no special treatment.

Remark 2. The single site Gibbs update for vertex i is the Markov kernel on Π given by

$$P_i(z, z') = \frac{p(z' \mid X)}{\sum_{z'' \in \Pi : T_i(z'') = T_i(z)} p(z'' \mid X)}\, \mathbf{1}\{T_i(z') = T_i(z)\}, \qquad z, z' \in \Pi,$$

that is, we keep $\rho = T_i(z)$ fixed and redraw c according to (1.3). If $T_i(z) = T_i(z')$, then both $p(z \mid X)P_i(z, z')$ and $p(z' \mid X)P_i(z', z)$ are equal to $p(z \mid X)p(z' \mid X)$ divided by the same normalizing sum, and otherwise both vanish. Hence P_i is reversible with respect to $p(\cdot \mid X)$. No coordinates, linear structure or topology on Π are needed for this.

Vertex i as the last vertex to arrive

For a permutation σ of $\{1, 2, \ldots, N\}$, write $\sigma(z) = \{\sigma(C) : C \in z\}$ and let X^σ be the adjacency matrix given by $X^\sigma_{\sigma(j),\sigma(l)} = X_{j,l}$. The CRP density (1.2) depends on z only through K and the block sizes, so $p(\sigma(z)) = p(z)$. Likewise, the density (1.1) depends on (X, z) only through the numbers of edges and non-edges between blocks, which do not change when the vertices of X and z are permuted together, so $p(X^\sigma \mid \sigma(z)) = p(X \mid z)$. Hence the posterior does not care about the order of the vertices, and in (1.3) we may pretend that i is the N-th vertex to arrive, even when $i = 1$. The computations below confirm this picture: the conditional distribution (1.3) factors as

$$p(z_i = c \mid z^{(-i)} = \rho, X) \propto [\text{CRP seating probability of the last vertex}] \times [\text{how well the edges of } i \text{ fit } c].$$

Prior part. For every block B of ρ, write $n_B = |B|$. Since $\Gamma(n_B+1) = n_B\Gamma(n_B)$ and $\Gamma(1)=1$, (1.2) gives

$$p(\rho \oplus_i B) = \frac{A^{K_\rho}\Gamma(A)}{\Gamma(A+N)} \prod_{B'\in\rho} \Gamma(n_{B'}) \cdot n_B, \qquad p(\rho \oplus_i \text{new}) = \frac{A^{K_\rho}\Gamma(A)}{\Gamma(A+N)} \prod_{B'\in\rho} \Gamma(n_{B'}) \cdot A,$$

where the second identity uses that $\rho \oplus_i \text{new}$ has $K_\rho + 1$ blocks. Hence

$$p(\rho \oplus_i c) \propto \begin{cases} n_B & \text{if } c = B \in \rho, \\ A & \text{if } c = \text{new}. \end{cases}$$

Since $\sum_{B\in\rho} n_B = N-1$, the normalized weights are $n_B/(A+N-1)$ and $A/(A+N-1)$, which are exactly the CRP probabilities for the N-th vertex to join block B or to create a new block.

Likelihood part. Write $X^{(-i)} = (X_{j,l})_{1\leqslant j<l\leqslant N,\ j,l\neq i}$ for the graph with vertex i removed and $x_i = (X_{i,j})_{j\neq i}$ for the edges of vertex i. For blocks B, B' of ρ (with $B = B'$ allowed), let $m_{B,B'}$ and $\bar{m}_{B,B'}$ be the numbers of edges and non-edges between B and B' in $X^{(-i)}$, where each unordered pair of distinct vertices is counted once (so that, when $B = B'$, these are the pairs of vertices inside B). For every block B' of ρ, let

$$r_{B'} = \sum_{j\in B'} X_{i,j}$$

be the number of edges between i and B', so that there are $n_{B'} - r_{B'}$ non-edges between i and B'. Finally, let

$$p(X^{(-i)} \mid \rho) = \prod_{\{B,B'\}} \frac{B(m_{B,B'}+a, \bar{m}_{B,B'}+b)}{B(a,b)},$$

where the product ranges over all unordered pairs $\{B, B'\}$ of blocks of ρ with $B = B'$ allowed; this is the right-hand side of (1.1) applied to the graph $X^{(-i)}$ and the partition ρ.

Proposition 3. For every block B of ρ,

$$p(X \mid \rho \oplus_i B) = p(X^{(-i)} \mid \rho) \prod_{B'\in\rho} \frac{B(m_{B,B'}+r_{B'}+a, \bar{m}_{B,B'}+n_{B'}-r_{B'}+b)}{B(m_{B,B'}+a, \bar{m}_{B,B'}+b)},$$

and

$$p(X \mid \rho \oplus_i \text{new}) = p(X^{(-i)} \mid \rho) \prod_{B'\in\rho} \frac{B(r_{B'}+a, n_{B'}-r_{B'}+b)}{B(a,b)}.$$

Proof. The blocks of $\rho \oplus_i B$ are $B\cup\{i\}$ and the blocks $B' \neq B$ of ρ. Pairs of vertices not involving i are counted exactly as in ρ. For every $B' \in \rho$, the $n_{B'}$ pairs $\{i,j\}$ with $j \in B'$ fall into the block pair $\{B\cup\{i\}, B'\}$ if $B' \neq B$, and into the block pair $\{B\cup\{i\}, B\cup\{i\}\}$ if $B' = B$; in both cases they add $r_{B'}$ edges and $n_{B'} - r_{B'}$ non-edges to the counts $m_{B,B'}$ and $\bar{m}_{B,B'}$. So, in (1.1), only the factors of the block pairs containing $B\cup\{i\}$ change, there is exactly one such factor for each $B' \in \rho$, and dividing each new factor by the corresponding factor of $p(X^{(-i)} \mid \rho)$ gives the first identity. The blocks of $\rho \oplus_i \text{new}$ are $\{i\}$ and the blocks of ρ. The block pairs of ρ keep their counts, the block pair $\{\{i\}, B'\}$ receives $r_{B'}$ edges and $n_{B'} - r_{B'}$ non-edges for every $B' \in \rho$, and the block pair $\{\{i\},\{i\}\}$ contains no pair of vertices, so its factor is $B(a,b)/B(a,b) = 1$. This gives the second identity. □

The factor $p(X^{(-i)} \mid \rho)$ is the same for every $c \in \mathcal{C}_\rho$ and therefore drops out of (1.3). Since the marginal distribution of $X^{(-i)}$ under $\rho \oplus_i c$ is the SBM with partition ρ (it only involves pairs of vertices not containing i), the remaining product is the conditional probability $p(x_i \mid X^{(-i)}, \rho \oplus_i c)$ of the edges of i given the rest of the graph. Each of its factors is a Beta–Bernoulli posterior predictive probability: in the hierarchical formulation of

Schmidt and Mørup (2013), the link probability $\theta_{B,B'}$ between blocks B and B' has prior Beta(a, b), its posterior given $X^{(-i)}$ is Beta$(m_{B,B'} + a, \bar{m}_{B,B'} + b)$, and

$$\frac{B(m_{B,B'} + r_{B'} + a, \bar{m}_{B,B'} + n_{B'} - r_{B'} + b)}{B(m_{B,B'} + a, \bar{m}_{B,B'} + b)}$$

is the probability, under this posterior, of the particular configuration of $r_{B'}$ edges and $n_{B'} - r_{B'}$ non-edges between i and B'. For a new block there is no previous observation, so the posterior is simply the prior Beta(a, b). These are exactly the quantities computed (on the logarithmic scale) in the code of Schmidt and Mørup (2013).

Conclusion. Combining the prior part with Proposition 3, for every $\rho \in \Pi^{(-i)}$ we obtain

$$p(z_i = c \mid z^{(-i)} = \rho, X) \propto \begin{cases} n_B \displaystyle\prod_{B' \in \rho} \frac{B(m_{B,B'} + r_{B'} + a, \bar{m}_{B,B'} + n_{B'} - r_{B'} + b)}{B(m_{B,B'} + a, \bar{m}_{B,B'} + b)} & \text{if } c = B \in \rho, \\ A \displaystyle\prod_{B' \in \rho} \frac{B(r_{B'} + a, n_{B'} - r_{B'} + b)}{B(a, b)} & \text{if } c = \text{new}, \end{cases} \tag{1.4}$$

where the proportionality constant is fixed by summing over the $K_\rho + 1$ options in $\mathcal{C}_\rho$. In words, $p(z_i \mid z^{(-i)}, X)$ is the probability that vertex i joins each block of ρ, or starts a new one, given the partition ρ of the other vertices: it is the CRP seating weight (n_B or A) times the posterior predictive likelihood of the edges of i under that choice, normalized over the $K_\rho + 1$ options. This description never uses $\mathbb{R}^N$ or block labels. After drawing c, we write the new state $\rho \oplus_i c$ in restricted growth string notation.

References

Charles Kemp, Thomas L. Griffiths, and Joshua B. Tenenbaum. Discovering latent classes in relational data. AI Memo 2004-019, Massachusetts Institute of Technology, Computer Science and Artificial Intelligence Laboratory, Cambridge, MA, September 2004. URL http://hdl.handle.net/1721.1/30489. Also issued as MIT-CSAIL-TR-2004-050.

Mikkel N. Schmidt and Morten Mørup. Nonparametric Bayesian modeling of complex networks: An introduction. *IEEE Signal Processing Magazine*, 30(3):110–128, May 2013. doi: 10.1109/MSP.2012.2235191.

Zhao Xu, Volker Tresp, Kai Yu, and Hans-Peter Kriegel. Infinite hidden relational models. In Rina Dechter and Thomas S. Richardson, editors, *Proceedings of the Twenty-Second Conference on Uncertainty in Artificial Intelligence (UAI 2006)*, pages 544–551, Arlington, VA, 2006. AUAI Press. URL https://arxiv.org/abs/1206.6864.